

Abdul Monaf Chowdhury

[Website](#) | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

University of Dhaka, Bangladesh

Email: monafabdul15@gmail.com

EDUCATION

Master of Science in Robotics and Mechatronics Engineering

August 2025 – Jan 2027

University of Dhaka, Bangladesh

Thesis: Value Propagation for Offline Goal Conditioned Reinforcement Learning

Supervisor: Dr. Md Mehedi Hasan

Bachelor of Science in Robotics and Mechatronics Engineering

Jan 2019 – Jan 2024

University of Dhaka, Bangladesh

CGPA: **3.87/4.00**, Ranked **2nd** place

Relevant Coursework:

Artificial Intelligence, Introduction to Machine Learning, Digital Image Processing and Robot Vision, Digital Signal Processing, Human-Robot Interaction, Advanced Robotics

Skills:

Software: C, C++, Python, MATLAB, PyTorch, JAX, Flax, TensorFlow, $\text{\LaTeX}$

Languages: English (fluent), Bangla (native)

RESEARCH INTERESTS

Embodied AI, Reinforcement Learning, Multi-modal Learning, Computer Vision, Vision Language Models

RESEARCH EXPERIENCE

Research Assistant

Oct 2025 – Present

AVIS Lab, University of Dhaka

Dhaka, Bangladesh

Supervisor: Dr. Md Mehedi Hasan

- Designed an efficient value learning algorithm extending the divide-and-conquer methods for offline goal-conditioned RL, and improved goal-reaching over the SOTA baselines on stochastic, deterministic, and stitching tasks in OGBench
- Evaluated goal-representation quality dependence on offline GCRL and developed Fourier positional encodings for state representation to improve downstream goal-reaching performance
- Two first-author papers are currently under review at **ICLR '27**

Research Assistant

Feb 2024 – Sep 2025

MAIM Lab, University of Dhaka

Dhaka, Bangladesh

Funding: Wellcome Leap (In Utero, California, USA)

PI: Dr. Niamh Nowlan, Co-PI: Dr. Abhishek Kumar Ghosh

- Collaborated with **University College Dublin** on the **Wellcome Leap In Utero** funded project titled “Translation of a Wearable Fetal Movement Monitor towards Stillbirth Prevention”
- Designed and implemented deep learning-based frameworks to analyze multimodal sensor signals from wearable belts to detect body movements, fetal kicks, and fetal hiccups
- Optimized signal processing of sensor data, presented hardware design feedback based on analytical findings, and validated hardware design changes
- Assisted in attaining **\$1M** funding extension, and eventually helped secure translational funding from Wellcome Leap to launch a **startup**

PROFESSIONAL EXPERIENCE

Lecturer

Mar 2026 – Apr 2026

Daffodil International University

Dhaka, Bangladesh

- Instructed undergraduate junior and senior year students on Introduction to AI, and Control Systems courses, and took relevant labs.

Instructor

National Camp, Bangladesh AI Olympiad

May 2025 – Aug 2025

Dhaka, Bangladesh

- Instructed national camp students on Unsupervised Learning, Deep Learning, and Computer Vision algorithms and architectures, took relevant labs, and illustrated Deep Learning evaluation strategies and techniques

PUBLICATIONS

- *LaGEA: Language Guided Embodied Agents for Robotic Manipulation* [Paper] [Code]
Abdul Monaf Chowdhury, AKM Moshir Rahman Mazumder, Rabeya Akter Fariya, Safaeid Hossain
43rd International Conference on Machine Learning, ICML '26
- *T3Time: Tri-Modal Time Series Forecasting via Adaptive Multi-Head Alignment and Residual Fusion* [Paper] [Code]
Abdul Monaf Chowdhury, Rabeya Akter Fariya, Safaeid Hossain
40th Annual AAAI Conference on Artificial Intelligence, AAAI '26
- *U-ActionNet: Dual-Pathway Fourier Networks with Region-of-Interest Module for Efficient Action Recognition in UAV Surveillance* [Paper]
Abdul Monaf Chowdhury, Ahsan Imran, Md Mehedi Hasan, Riad Ahmed, AKM Azad, Salem A. Alyami
IEEE Access, 2024. IF - 3.4
- *FFT-UAVNet: FFT Based Human Action Recognition for Drone Surveillance System* [Paper]
Abdul Monaf Chowdhury, Ahsan Imran, Md Mehedi Hasan
5th IEEE International Conference on Sustainable Technologies for Industry 5.0 (STI), 2023

MANUSCRIPT SUBMITTED

- *GTRL: Grounding Divide-and-Conquer Value Learning with Temporal Differences* [Paper] [Code]
Abdul Monaf Chowdhury, MD Sameer Iqbal Chowdhury, Shifat E. Arman, Md Mehedi Hasan
International Conference on Learning Representations - ICLR 2027
- *Do Better Goal Representations Improve Goal-Conditioned Reinforcement Learning?* [Paper] [Code]
Syed Nazmus Sakib*, **Abdul Monaf Chowdhury***, Nafiul Haque, Shifat E. Arman, Md Mehedi Hasan
International Conference on Learning Representations - ICLR 2027
* = Equal-Contribution
- *Kernelized Parameter Sensitivity Atlases for Amortized Conditional Interpretability*
Davide Wiest, Ahanaf Hasan Ariq, **Abdul Monaf Chowdhury**
Thirtieth Annual Conference on Artificial Intelligence and Statistics - AISTATS 2027
- *Certifying Whether to Adapt: Baseline-Relative PAC-Bayes-MMD Guarantees for Test-Time Adaptation*
Ahanaf Hasan Ariq, Hasaan Ahmad, Aayush Parekh, **Abdul Monaf Chowdhury**
Thirtieth Annual Conference on Artificial Intelligence and Statistics - AISTATS 2027
- *Counting Through Occlusion: Framework for Open World Amodal Counting* [Paper] [Code]
Safaeid Hossain, Rabeya Akter Fariya, **Abdul Monaf Chowdhury**, Md Mehedi Hasan, Md. Jubair Ahmed Surov
Winter Conference on Applications of Computer Vision - WACV 2027

RESEARCH PROJECT

Offline Goal-Conditioned Reinforcement Learning [Code] [Paper]

Jan 2026 – Sep 2026

Reinforcement Learning, Representation Learning

- Designed an efficient value learning algorithm extending the divide-and-conquer methods for offline goal-conditioned RL, and improved goal-reaching over the SOTA baselines on stochastic, deterministic, and stitching tasks in OGBench
- Evaluated goal-representation quality dependence on offline GCRL and developed Fourier positional encodings for state representation to improve downstream goal-reaching performance

Open World Amodal Counting [Code] [Paper]

Jan 2026 – Apr 2026

Multimodal Learning, Vision Language Model, Representation Learning

- Architected a hierarchical Feature Reconstruction Module fusing visible context with vision-language priors to reconstruct occluded class-discriminative features, anchored by teacher-student Visual-Equivalence consistency
- Built occlusion-augmented FSC-147 and CARPK benchmarks; achieved SOTA amodal counting from RGB alone, reducing MAE by **32.2%** (FSC-147), **40.7%** (CARPK), and **40.0%** (CAPTURE-Real, cross-dataset)

Language Guided Embodied Agents [\[Code\]](#) [\[Paper\]](#)

Aug 2025 – Jan 2026

Embodied AI, Vision Language Model, Reinforcement Learning

- Designed a **Qwen2.5-VL-3B**-driven “episodic self-reflection” module that generates natural-language self-assessments of each trial, pinpointing successes and failure causes
- Fused self-reflections with CLIP-style vision–language feedback and goal images into a dense, semantically grounded reward for **Soft Actor-Critic**, accelerating convergence on Meta-World manipulation tasks

Tri-Modal Time Series Forecasting [\[Code\]](#) [\[Paper\]](#)

Apr 2025 – Aug 2025

Large Language Model, Deep Learning, Signal Processing

- Architected an Adaptive Dynamic Multi-Head Cross-Modal Attention module with channel-wise residual skip-connections for fine-grained alignment of temporal and auxiliary features
- Engineered an **FFT**-based frequency-domain encoder and a Rich-Horizon Gating Fusion of spectral and temporal encodings; beat **SOTA** on multivariate time-series forecasting

Human Action Recognition from Drone footage [\[Paper\]](#)

Jan 2023 – Jan 2024

Computer Vision, Deep Learning

- Designed a hybrid model combining a modified 3D CNN with an FFT-based action recognition module to detect suspicious human activity in UAV surveillance
- Built a lightweight MobileNetV2 + BiLSTM pipeline human activity detection for edge devices, reducing inference time

Proxemics & Social Interaction Patterns in ASD Children

Sep 2023 – Jan 2024

Human-Robot Interaction, Deep Learning

- Formulated a **YOLOv8**-based system to determine the ideal proxemics of children with autism spectrum disorder (ASD) interacting with a **NAO** robot
- Analyzed the behavioural responses of 20 children with ASD to specific NAO robot actions

AWARDS & SCHOLARSHIPS

- Dean’s Award for Best Undergraduate Result, University of Dhaka, 2024
- Engineering Faculty Undergraduate Merit Scholarship, University of Dhaka, 2024
- 5th, Dataverse Challenge - ITVerse, Bangladesh, 2023; [\[Report\]](#)
- 2nd, Intra-Department Soccer Bot Championship, University of Dhaka, 2019
- Sylhet Board Scholarship, Higher Secondary Certificate Examination 2018

WORKSHOP/CONFERENCE ATTENDED

- 5th International Conference on Sustainable Technologies for Industry 5.0 (STI), Dhaka
- 40th Annual AAAI Conference on Artificial Intelligence, AAAI ’26
- 43rd International Conference on Machine Learning, ICML ’26

REVIEWER

- AAAI 27 **Program Committee**, AAAI 26, ICML 26, ICLR 27, IEEE Access

LEADERSHIP/VOLUNTEER ACTIVITIES

General Secretary

Mar 2022 – Feb 2024

RMEDU Student Club, University of Dhaka

Academic Team Mentor

Sep 2019 – Aug 2022

Bangladesh Robot Olympiad

Program Co-Ordinator

Jul 2021 – Jun 2022

IEEE Robotics & Automation Society, University of Dhaka